

not obvious to a person skilled in the art. In view of the negative definition in Art. 56 EPC, the board considered that inventive step could never in fact be proven but that secondary indicia might sometimes show it was plausible. A lack of inventive step by contrast could actually be proven by conclusively showing that a solution was obvious. So proving a lack of inventive step was possible, in principle, and indeed necessary where such a lack was alleged; it was then not enough to point to secondary indicia.

In T 179/18 the board was not persuaded that any of the secondary indicia claimed by the appellant conferred inventiveness on the subject matter of claim 1. It held that secondary indicia such as the satisfaction of a long-felt need, the overcoming of a prejudice in the professional world, or economic success, were auxiliary considerations in assessing inventive step. However, in the case in hand, application of the problem-solution approach led to the clear result that the subject-matter of claim 1 did not involve an inventive step.

10.2. Technical prejudice

According to the case law of the boards of appeal (see T 119/82, OJ 1984, 217; T 48/86), inventiveness can sometimes be established by demonstrating that a known prejudice, i.e. a widely held but incorrect opinion of a technical fact, needs to be overcome. In such cases, the burden is on the patentee (or patent applicant) to demonstrate, for example by reference to suitable technical literature, that the alleged prejudice really existed (T 60/82, T 631/89, T 695/90, T 1212/01). A high standard of proof to demonstrate prejudice is required (T 1989/08).

A prejudice in any particular field relates to an opinion or preconceived idea widely or universally held by experts in that field. The existence of such prejudice is normally demonstrated by reference to the literature or to encyclopaedias published before the priority date. The prejudice must have existed at the priority date, any prejudice which might have developed later is of no concern in the assessment of inventive step (T 341/94, T 531/95, T 452/96, T 1212/01, T 25/09, T 99/19).

Generally speaking, prejudice **cannot** be demonstrated by a statement in a single patent specification (T 19/81, OJ 1982, 51; T 392/88; T 900/95), since the technical information in a patent specification or a scientific article might be based on special premises or on the personal view of the author. However, this principle does not apply to explanations in a standard work or textbook representing common expert knowledge in the field concerned (T 321/87; T 392/88; T 453/92; T 1212/01).

In T 515/91 the board regarded "ABC Naturwissenschaft und Technik" as a standard work.

In T 943/92 the existence of a prejudice was supported by a specialist book which reflected the technical knowledge in the special field of the contested patent. This book did not contain the opinion of just one specialist author, but that of experts in the field, as it had resulted from the collaboration of "numerous recognised scientists, technicians and practitioners as well as associations and institutes". General critical remarks in one textbook were not sufficient to substantiate an alleged prejudice, if a plurality of prior art documents pointed to the opposite (T 134/93).

Generally speaking, established board of appeal case law is **very strict** when it comes to recognising the existence of a prejudice. For a prejudice to be considered actually to exist, there must be a prevailing but incorrect school of thought widespread throughout the entire technical field. A solution put forward as overcoming a prejudice must clash with the prevailing teaching of experts in the field, i.e. their unanimous experience and notions, rather than merely cite its rejection by individual specialists or firms (T 531/95; see also T 62/82, T 410/87, T 500/88, T 74/90, T 793/97, T 2453/09).

In T 2044/09 the board did not interpret a declaration by a technical expert in the field as evidence that there was a prejudice in the prior art at the priority date; rather, this declaration was taken as the opinion of one expert, issued almost ten years after the priority date.

In T 69/83 (OJ 1984, 357) the board held that the mere fact that the disadvantage resulting from the omission of a component of a mixture recognised in the art as advantageous was accepted suggested conscious acceptance of a disadvantage, not the overcoming of a prejudice. fact that a disadvantage is accepted or the prejudice simply ignored does not mean that a prejudice has been overcome (see also T 262/87; T 862/91).

The board in T 1989/08 observed that this meant the standard of proof was almost as high as that required for common general knowledge in the art. For example, it was not enough that the opinion or idea was held by a limited number of individuals or that it was a prevalent view within a given firm, however large. In the case in hand all the evidence offered in support of the alleged prejudice amounted to no more than ten documents, all of which were either specialist papers or patents. This small number of publications intended for a select readership in the field was in itself a tenuous basis for asserting prejudice (see also T 25/09).

In T 179/18 the board considered that the three years separating the filing date of the patent and the filing date of the disclosed prior art was not a long period indicative of a technical prejudice.

In T 1212/01 the patent related to pyrazolopyrimidinones for the treatment of impotence (Viagra). The patentee referred to some thirty scientific articles in order to demonstrate the existence of a technical prejudice that drugs lowering blood pressure were a cause of impotence rather than a form of treatment for that condition. The board, however, stated that the contents of such a selection from the prior art could not be considered per se as creating a technical prejudice against oral treatment of male erectile dysfunction. Such a prejudice could only be established by proving that, in relation to the technical solution, a relatively widespread error or misapprehension about the technical invention existed among skilled workers in the relevant field before the priority date of the patent in suit. This was not the situation in the case in hand.

In T 550/97 the respondent (patent proprietor) had argued that, years after the invention, technically less advanced solutions had been filed for and marketed as a means of integrating different mobile radio networks. However, the board did not consider the fact that arguably less advanced solutions had been developed later, to be evidence of

inventive step, as there was no reason to assume that later development was attributable to a technical prejudice which the invention had needed to overcome.

In T 347/92 the board pointed out that the finding of a relatively small operating window in an area which, according to the teaching of the most recent publications, was considered inaccessible, could not be considered obvious to a person skilled in the art.

In T 984/15 the board found that the fact that something was not explicitly disclosed, or even that it was explicitly prohibited, in a technical specification must not be necessarily interpreted as a prejudice of technical nature. There might be other non-technical reasons, such as circumventing a patented technology or simply lack of time in the meetings for further discussions that led the drafting group to dismiss or ignore an otherwise valid technical option.

One form of secondary indicia in the nature of a "technical prejudice" is a **development of the art in a different direction** (T 24/81, OJ 1983, 133; T 650/90; T 330/92).

In T 883/03 the board found that the teaching that could have led to the characterising feature of claim 1 had long formed part of the state of the art; yet for all that time experts had been "**blind**" to that knowledge. In the case in point that was a further indication of the inventiveness of the solution proposed in claim 1.

In T 872/98 the board pointed out that the presence of secondary indicia might also be attested by the fact that a competitor had, shortly after the priority date, filed a patent application with the German Patent Office in which the invention took an entirely different direction to the European application.

In T 779/02 the board indicated that a prejudice could be proved by the fact that the closest prior art and the invention were separated by a long period of time (over 16 years in the case in hand), during which time the only solutions pursued led away from the invention, and the solution provided by the invention only becoming acceptable to experts in the field after this time.

10.3. Age of documents – time factor

The age of documents known long before the filing date might only be an indication of an inventive step if a need for the solution of an unsolved problem had existed for the entire period between the date of the documents and that of the invention (T 79/82, T 295/94). Nevertheless, the long period of time to be considered was not the period that had elapsed between the publication of a document and the filing of the European patent application disclosing the teaching of that document, but that between the time the problem became apparent and the date of filing of the European patent application providing a solution (T 478/91).

In T 273/92 the board held that a period of 23 years between the publication date of the document deemed to be the closest prior art and the priority date of the contested patent in an economically significant and frequently studied field could normally be viewed as an

indication of the presence of inventive step. In T. 203/93 and T. 795/93 a period of 11 years was considered to be an indication in support of inventive step, in T. 986/92 a period of 70 years, in T. 478/91 80 years and in T. 626/96 60 years. See also T. 774/89, T. 540/92, T. 957/92, T. 697/94, T. 322/95, T. 255/97, T. 970/97, T. 6/02, T. 2271/08.

In T. 330/92 the documents reflecting the general knowledge available to experts in the field of the application (injection mould technology for cashcard holders) had been published at least 17 years before the filing date of the contested patent. The board of appeal pointed out that the elements which could have led to the feature combination of claim 1 had thus long been known in the prior art for some time. Nevertheless the experts had for all this time been "blind" to these findings. Nor had other applicants in the same field made use of the knowledge in question.

In T. 1077/92 the board faced the unusual situation of a problem and its ready solution having co-existed for 100 years in general, and more recently in a field of intensive research, and still the seemingly obvious step had not been taken. The board concluded that, as no other explanation could be found, this must have been because inventive insight was needed. In T. 617/91 the board compared the 100 year interval in T. 1077/92 to the seven year interval between D1 and the priority date of the alleged invention in the case in hand. The board noted that despite there being a strong incentive amongst SG iron makers throughout that time interval to look for better methods which would be safer, and more economical, throughout that interval the teaching of D1 was ignored. Consequently, although D1 seen in hindsight seemed to suggest a solution, the board drew the inference that the generality of its teaching was such that it remained unnoticed, and would have remained so but for the inventive insight of the appellant.

In T. 123/97 the board found the failure to adopt an obvious solution to the technical problem underlying the patent in suit may have resulted from a variety of causes; for example, there may have been a commercial reason for not adopting this new technique, because the old technique was found satisfactory by the clients and could also be improved, thus avoiding considerable investment costs involved in the adoption of a new technique on an industrial scale. Therefore in the board's judgement, the claimed use according to claim 1 did not involve an inventive step.

The board in T. 833/99 found that, in addition to the other factors it had mentioned in examining whether a method of producing a grooved-rail frog for tramlines was inventive, it also had to be taken into account that citation D1 dated from 1930 and that, in the 60 years between it and the invention at issue, no skilled person had had the idea of taking up what it suggested and applying the method described to a single-piece frog. That was a serious indication of non-obviousness which could not be ignored.

The board in T. 1192/09, having found that there was an inventive step, added that the actual developments in the relevant technical field, as evidenced by the documents on file, provided additional support in favour of an inventive step. The board noted in this respect that D8 (closest prior art) had been published about 12 years before the filing date of the patent in suit. The fact that the very inventors of D8 had not proposed the claimed structure, with its uncontested advantages, until 12 years after the publication of D8, and

that this structure had later been praised in D10 (European patent, same inventors as D8, underlying application filed two months after the patent in suit) and made the subject of a granted European patent was considered by the board to be an additional indication of the non-obviousness of the concept which formed the basis of the claimed invention.

10.4. Satisfaction of a long-felt need

The fact that the state of the art has been inactive over a long period prior to the invention may be an indication that an inventive step is involved if during that time an urgent need for improvement has demonstrably existed (see T.109/82, OJ 1984, 473; T.555/91; T.699/91). This indication is closely linked to the positive indications of the time factor and age of the cited document.

In T.605/91 the board stated that it was not sufficient if only one individual skilled person discovered a "long-felt need". Only if various and repeated attempts to deal with the relevant disadvantages could be identified would such a long-felt need appear to have persisted.

In T.1014/92 the board did not accept the appellant's further argument that the long period of time (about 35 years) during which documents (1) and (2) had been available to the public without having been combined, was in itself cogent evidence that there was no obvious connection between them. The board held that this conclusion might only be drawn if evidence relating to time were corroborated by other evidence, such as long-felt want (see also T.1183/06).

In T.271/84 (OJ 1987, 405) the board found that where a process had been performed successfully on a commercial scale for more than 20 years in spite of economic disadvantages associated with it, and the claimed invention provided a solution to the technical problem of avoiding such economic disadvantages, this supported a finding of inventive step.

10.5. Commercial success

In principle, commercial success alone is not to be regarded as indicative of inventive step, but evidence of immediate commercial success when coupled with evidence of a long-felt want is of relevance provided the commercial success derives from the technical features of the invention and not from other influences (e.g. selling techniques or advertising). See Guidelines G-VII, 10.3 – March 2022 version.

In T.106/84 (OJ 1985, 132) the board found that the appellant's assertion that the commercial success his device enjoyed was merely based on the superior performance of the product related to the features claimed, rather than extraordinary sales promotion efforts, must be considered in the light of the fact that the appellant's company was rather small and conceivably could not afford to indulge in major sales campaigns and sophisticated marketing techniques. The board was thus prepared to accept that the commercial success of the invention stemmed from the technical advantages related to the features claimed.